

Efficient Retrieval for Local Language Models

1. Introduction

Retrieval augmented generation combines a document index with a language model so that answers can be grounded in source text. This paper studies retrieval strategies suited to small, locally hosted language models running under tight memory budgets.

2. Method

We combine dense retrieval with a sparse BM25 index and merge the two ranked lists using reciprocal rank fusion. Reciprocal rank fusion assigns each document a score equal to the sum of one over the rank of that document across the contributing ranked lists.

The fused ranking is optionally refined with a cross-encoder reranker before the top results are passed to the language model.

3. Results

On a held-out set of technical documentation questions, the fused retrieval pipeline improved recall at five over dense-only retrieval by a wide margin, while keeping memory usage low enough to run entirely on consumer hardware without a discrete GPU.